

Commercial VRP vs. Disaster Relief VRP: Key Differences

Dimension	Commercial VRP	Disaster Relief VRP
Primary Objective	Minimise cost / distance	Minimise suffering / response time
Fleet Availability	Known, controlled	Uncertain, may be damaged
Demand	Known or forecast	Unknown / rapidly evolving
Road Network	Reliable	May be blocked or destroyed
Time Horizon	Planned in advance	Real-time, dynamic re-routing
Objectives	Single (cost)	Multiple (equity, speed, coverage)
Vehicle Type	Homogeneous fleets	Diverse (trucks, helicopters, boats)
Priority	Not typically considered	Life-saving priority is critical
Return to Depot	Required each trip	May not return; one-way routes
Uncertainty	Low	Extremely high